

Himanshu Chavan

GitHub : Himanshujchavan | LinkedIn : chavan-himanshu | Email : chavanhimanshu37@gmail.com

Professional Summary

Computer Science undergraduate with a comprehensive background spanning Machine Learning, robust backend architecture, and full-stack development. Proficient in engineering intelligent multi-agent systems using LangChain, alongside designing scalable microservices and asynchronous REST APIs using FastAPI, Python, and Node.js. Experienced in building responsive user interfaces with React/Next.js, managing relational and NoSQL databases, and containerizing application workflows via Docker. Maintaining an active lifestyle through a passion for playing volleyball and traveling.

Education

Yeshwantrao Chavan College of Engineering (YCCE), Nagpur

B.Tech in Computer Science and Engineering

2023 – 2027

CGPA: 7.83/10.0

Projects

Real-Time Fraud Detection System

- Developed a real-time fraud detection platform using FastAPI, Python, and a microservices architecture.
- Built anomaly detection and risk-scoring models using XGBoost, Random Forest, and behavioral analytics.
- Engineered transaction, device, and geolocation features to identify suspicious activities and fraud patterns.
- Designed scalable REST APIs and containerized services with Docker for real-time model inference.

AI Multi-Agent Preventive Healthcare Platform

- Developed an AI-powered healthcare platform for diabetes risk assessment and clinical decision support using LangChain and FastAPI.
- Orchestrated specialized AI agents for medical report analysis, risk prediction, and emergency alerting.
- Built medical PDF extraction pipelines for structured healthcare analytics.
- Implemented secure, asynchronous backend services with JWT authentication and resilient recovery mechanisms.

VA-2 Genomic Data Analysis Platform

- Developed a blockchain-enabled genomic data analysis platform using FastAPI and PostgreSQL.
- Built genomic data validation and preprocessing pipelines for variant analysis and fine-tuned the **Evo 2** genomic foundation model for domain-specific predictions.
- Designed scalable database schemas to support genomic research workflows.
- Integrated NFT minting on BNB Smart Chain for decentralized ownership and verification of research data.

Technical Skills

Languages	Python, C++, JavaScript
AI/ML	NumPy, Pandas, TensorFlow, PyTorch, Scikit-learn, LangChain, LLM Integration, Multi-Agent Systems
Web & Backend	FastAPI, React, Next.js, Node.js, REST APIs, JWT Auth, Postman
Databases	PostgreSQL, MongoDB, SQL
DevOps & Cloud	Docker, AWS (EC2, S3), Streamlit

Leadership & Achievements

- **Secretary**, ACM Student Chapter; previously served as **Webmaster**, managing the website and technical events.
- Earned **Honorable Mention (4th Place)** in the **DSCI Track** at the **BNB Hackathon**.
- Secured **5th Place** among **40 finalist teams** in a **Cyber Hackathon** organized by the **Nagpur Police Commissionerate**.